Week 4 – Customer Segmentation Deep-Dive

Behavioral segmentation using K-Means clustering

Project deliverable | Customer profiles | 10 November 2026

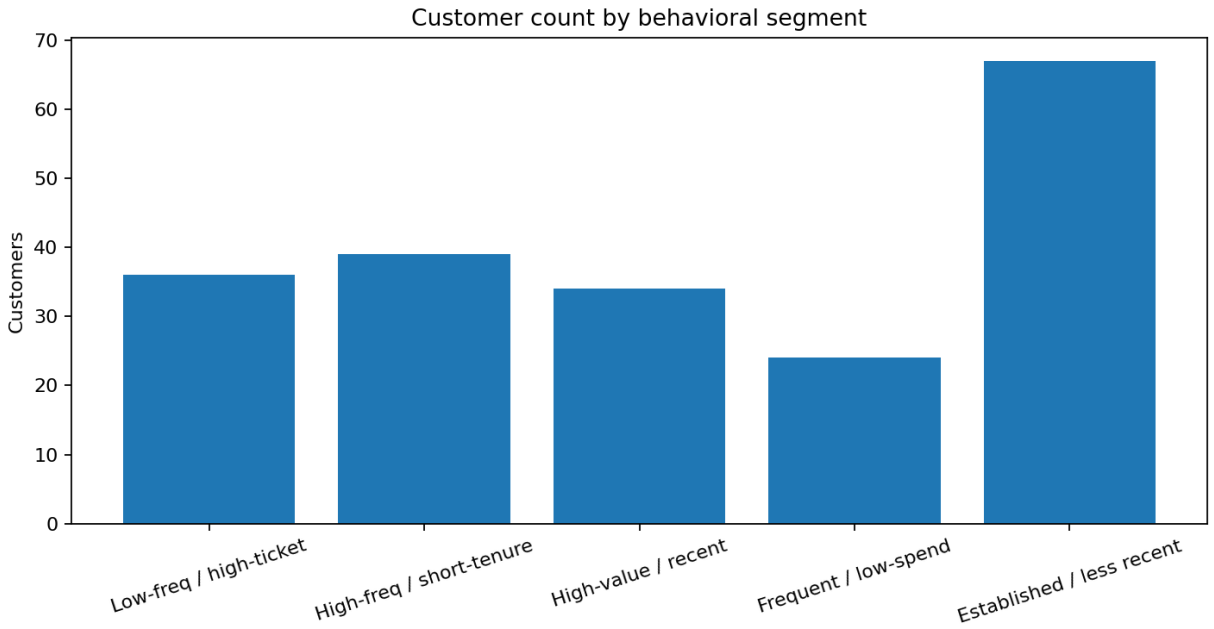
Executive summary

This analysis segments 200 customers into 5 behavioral groups using purchase frequency, cumulative revenue, average order value, purchase tenure, and relative recency. The selected K-Means solution achieved a silhouette score of 0.3223 among tested values k=2–5, indicating moderate separation with some overlap. The customer extract contains 6 churn flags (3.0%); all six appear in Segment 1. Churn is used only to describe the resulting segments and is not included as a clustering feature.

Customers	200	**Recorded revenue**	4,992,621 source units
Average revenue/customer	24,963.10	**Churn flags**	6 (3.0%)
Selected clusters	5	**Silhouette score**	0.3223
Recency reference date	2026-07-07	**Data grain**	One row per customer

Segment overview

Segment	Customers	Avg. orders	Avg. revenue	Avg. recency (days)	Churn rate
S1: Low-frequency, high-ticket / churn-risk	36	3.1	25,513	235	16.7%
S2: High-frequency, short-tenure anomaly	39	14.3	25,128	237	0.0%
S3: High-value, recently active	34	10.1	32,762	29	0.0%
S4: Frequent, low-spend	24	14.2	4,136	205	0.0%
S5: Established, mid-value / less recent	67	14.2	28,074	200	0.0%



Methodology

1. Analytical choice

Customer segmentation was selected because the customer table contains customer-level aggregate behavior. A churn prediction model would be unreliable with only six positive churn labels (3%). A full cohort-retention analysis is not supported because transaction-level monthly activity is not provided.

2. Feature engineering

Total_Orders, Total_Revenue, Avg_Order_Value = Total_Revenue / Total_Orders, Tenure_Days = Last_Purchase_Date – First_Purchase_Date, and Recency_Days = latest observed Last_Purchase_Date – each customer's Last_Purchase_Date. The snapshot date is the maximum observed last-purchase date: 2026-07-07.

3. Preprocessing

Numeric features were median-imputed where necessary, transformed using log1p to reduce skew, and standardized with StandardScaler. K-Means was fitted with random_state=42 and n_init=20.

4. Model selection

K-Means was evaluated at k=2, 3, 4, and 5 using silhouette score. k=5 was selected because it had the highest score (0.3223) among the tested candidates. This is a heuristic selection, not proof that five groups are objectively true.

5. Leakage control

Churned, Gender, Age_Band, City, and Acquisition_Channel were excluded from the clustering features. Churn and profile attributes were used only after fitting to help describe segments.

Model selection results

Clusters (k)	Silhouette	Smallest cluster
2	0.2303	84
3	0.2609	39
4	0.3084	23
5	0.3223	24



Interpretation and recommendations

Segment 1 – Low-frequency, high-ticket / churn-risk

36 customers; average 3.11 orders; average revenue 25,513; churn rate 16.67%. All six churn flags fall in this segment. Prioritize personalized re-engagement and validate whether a small number of high-value orders drive the average. The churn rate is descriptive and based on six customers only.

Segment 2 – High-frequency, short-tenure anomaly

39 customers; average 14.28 orders; average tenure 1.18 days. The short first-to-last purchase span is unusual given the order count. Validate date semantics, duplicate/order aggregation rules, and source-system logic before targeting.

Segment 3 – High-value, recently active

34 customers; average revenue 32,762; average recency 29 days. Protect engagement with loyalty benefits, relevant cross-sell, and service-quality monitoring.

Segment 4 – Frequent, low-spend

24 customers; average 14.21 orders but average revenue 4,136. Test bundles or basket-building offers with margin safeguards.

Segment 5 – Established, mid-value / less recent

67 customers; average revenue 28,074; average recency 200 days. Consider a measured win-back campaign and monitor response.

Assumptions and limitations

• Currency/unit for Total_Revenue is not specified; values are reported in source units.

• Recency is relative to the latest date in this extract, not to today's date. It cannot be interpreted as current days-since-purchase.

• Tenure is the interval between first and last purchase dates; it is not account age.

• The dataset is aggregated at customer level. It does not contain transaction-level timestamps needed for true cohort retention curves or purchase sequence analysis.

• Only six churn labels are present, so segment-level churn differences are not statistically conclusive. No causal claims are made.

• Silhouette score 0.3223 indicates moderate/overlapping clusters. Results are exploratory and should be validated with business stakeholders and a later data refresh.

• Property listings and product catalogue are retained as project context but not joined to customers because no common relationship key was provided.

Reproducibility and deliverables

Run `python src/run_analysis.py` after installing `requirements.txt`. The script regenerates customer segment assignments, segment profiles, model-selection scores, and supporting age-band/acquisition-channel profiles. See README.md and reports/methodology_and_assumptions.md for implementation details.